

Active Silence: Designing Temporal Idle States for Calm Engagement in Public Interactive Displays

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Abstract

Public interactive displays often treat non-use as failure, defaulting to black screens, hard resets, or attention-grabbing loops. Such designs can increase ambient distraction and intensify visitors' pressure to perform immediately. We present the ambient installation, an exhibition-deployed generative audiovisual display, and investigate idle time as a designed interaction state. Inspired by purposeful blankness and lingering traces in visual aesthetics, we introduce Active Silence: when a camera-based presence proxy indicates no nearby participant, the system does not reset. Instead, it transitions into a low-activity sedimentation mode where traces decay with a long half-life and audio becomes sparse, sustaining a calm, watchable state that invites gradual entry. We conducted an exhibition study comparing Active Silence with a conventional black-screen idle using structured observation and brief intercept interviews. Results suggest that temporal idle "textures" support spectating, increase gentle transitions from watching to engaging, and reduce perceived pressure to immediately perform for the display.

CCS Concepts

• **Human-centered computing** → **Interaction design theory, concepts and paradigms**; Field studies.

Keywords

Idle states, Temporality, Public displays, Presence sensing, Observational study, Psychological amenity.

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1 Introduction

Idle time in public interactive displays is typically treated as a problem to be eliminated: a black screen signals inactivity, a hard reset abruptly clears all visual evidence of recent user interactions, and an attract loop demands immediate attention. While these strategies are operationally convenient, they also act as social signals in shared spaces. A display that looks inert can discourage lingering;

a display that shouts can create ambient distraction and raise the perceived stakes of participation.

HCI has long argued for calm and ambient technologies that respect peripheral attention and integrate aesthetically into everyday settings [11, 15]. Public display research similarly shows that engagement is often gradual, moving from implicit awareness to explicit interaction and from public observation to personal involvement [10, 14]. Yet, the idle state itself remains under-specified as an interaction state with temporal character, despite the fact that it is frequently the first thing passersby encounter. Treating idle as merely a timeout overlooks its role in setting tone, clarifying permission to approach, and supporting spectatorship as a legitimate mode of engagement [4].

We propose that idle can be designed as *temporal texture*: perceivable time-based qualities such as decay, persistence, and sparsity that shape whether the display feels demanding or permissive. We introduce **Active Silence**, an idle policy in which absence does not trigger a reset. Instead, the system enters a sedimentation mode that slowly decays traces and thins audio, maintaining a calm, watchable presence that invites gradual entry. We evaluate Active Silence in an exhibition deployment against a conventional black-screen/reset idle using structured observation and short intercept interviews. Beyond reporting a system, we contribute a reusable vocabulary for describing idle textures and a set of implications for calm public interaction and ethical sensing.

This paper closes with a practical claim: designing for idleness is designing public etiquette. Treating the idle state as a first-class design material can reduce perceived pressure while still supporting engagement.

2 Related Work

Public displays sit at the intersection of ambient information, social interaction, and non-use. This section connects these strands to motivate why idle states deserve deliberate design.

Ambient display research frames effectiveness in terms of peripheral awareness and aesthetic integration, while also documenting that evaluation is difficult and often requires in-situ deployment [9, 11]. Early domestic and long-term ambient systems show that impact can accumulate through familiarity rather than immediate task completion [5]. These perspectives motivate our focus on how a display behaves when unattended, because unattended time is where an ambient system most clearly signals whether it belongs in a space.

Public display work has emphasized gradual engagement. Vogel and Balakrishnan describe how interactive public ambient displays can support transitions from implicit to explicit interaction using



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contextual cues and lightweight gestures [14]. Multi-touch installations in exhibitions highlight spectatorship and social learning as central mechanisms of discovery [10]. Other work shows how displays can guide users' positioning, making clear that approach and stance are part of interaction and can be designed for [1]. These accounts imply that the unattended state should not be treated as outside the interaction: it is a major determinant of whether spectators become participants.

Non-use scholarship, alongside prior literature exploring the potential benefits of embracing inactive displays [6], argues that not interacting is meaningful and socially shaped, not a failure mode [3]. This matters in public space where people may avoid interaction because of perceived surveillance or uncertainty about norms. Slow technology and temporality oriented design treat time as a material for reflection and revisitation, suggesting that persistence and decay can be deliberately crafted rather than avoided [8, 12, 13]. Active Silence builds on these lenses by applying temporality to a concrete but under-discussed surface: the idle policy of a public interactive display.

In summary, prior work suggests that calmness, gradual entry, and meaningful non-use are compatible goals. The missing piece is an actionable account of idle as temporal design in public space, evaluated in a real deployment.

3 System and Design: Active Silence

This section introduces the installation and the core mechanism. The design intentionally limits sensing to keep the contribution about temporality rather than personal inference. The installation is a public display with speakers showing generative visuals and sparse generative audio. Touch interaction increases motion and saturation, leaving visible traces that would normally be cleared by a reset. The work was deployed in a gallery-like exhibition setting where passersby could stop, watch, and interact without instruction beyond a small "touch to explore" label.

3.1 Presence Proxy (Binary, Non-Identifying)

Active Silence relies on a camera-based presence proxy (zone motion threshold) that outputs a binary state: present when someone is near the interaction zone and absent otherwise. The system does not perform face recognition, demographic inference, or affect classification; it processes locally and stores no images. This choice reduces risk and aligns with the study focus on how *unattended* behavior shapes entry rather than how personal profiles tailor content.

3.2 From Reset to Sedimentation: Temporal Texture

In the baseline condition, absence triggers a fade to black followed by a full reset. In Active Silence, absence triggers a *sedimentation mode*: the display remains on but becomes quieter in both vision and sound. Instead of clearing, traces decay slowly; motion becomes minimal; audio events become rare. When someone returns, new interaction grows from what is already there, making re-entry continuous rather than performative.

We treat sedimentation as a small design space with three temporal parameters. *Half-life* controls how quickly traces fade; *persistence*

controls what remains after surface detail decays; *audio sparsity* controls the spacing of sonic events. Rather than prescribing universal values, we present these as tunable properties that can match local norms of quietness, crowding, and attentional load.

A brief implementation note matters for later discussion: Active Silence can be paired with reduced brightness and sparser rendering updates during idle, keeping computation and luminance lower while avoiding the social signal of "off". This is not a claim that any-on is always greener, but a claim that calm invitation and energy-aware operation need not be opposites.

To conclude this section, Active Silence is a simple shift in policy with outsized experiential consequences: it makes *unattended* behavior continuous, legible, and calm, reframing idle as the first interaction.

4 Method

We evaluated Active Silence in an exhibition context. Because behavior alone can be ambiguous in public space, we combined structured observation with short intercept interviews to ground interpretations about invitation and pressure.

4.1 Design and Procedure

We compared two idle policies in the same installation and location with identical hardware and content parameters: (A) black-screen/reset idle and (B) Active Silence. Conditions alternated in time blocks across exhibition days and were counterbalanced to reduce time-of-day effects. For each block we recorded contextual notes such as crowding and nearby events to help interpret changes that were likely unrelated to idle policy.

4.2 Structured Observation

A researcher conducted structured observation using a coding sheet and stopwatch. We operationalized engagement as a sequence commonly described in public display settings: pass-by, pause, spectate, approach, touch interaction, and leaving [4, 14]. For each observed visitor or group, we recorded whether they approached into the interaction zone, how long they dwelled after an initial pause, whether spectating transitioned to touching, whether they re-approached later within the same block, and whether they exhibited repeated approach/retreat hesitation. When groups approached together, we coded them as a single episode and noted whether any member initiated touch.

Given the realities of evaluating ambient systems in the wild [9], across blocks we observed 4 hours; we treat these measures descriptively and use them primarily to compare patterns across conditions rather than to make population-level claims.

4.3 Intercept Interviews

To avoid projecting internal states onto observed behavior, we conducted brief intercept interviews with **N=10** visitors (five per condition) immediately after they moved away from the display. Interviews lasted roughly 1–2 minutes, were voluntary, and were designed to minimally disrupt the exhibition experience.

Questions focused on what made them stop or keep walking, whether the display felt inviting or demanding and why, whether they felt pressure to do something immediately, and how they

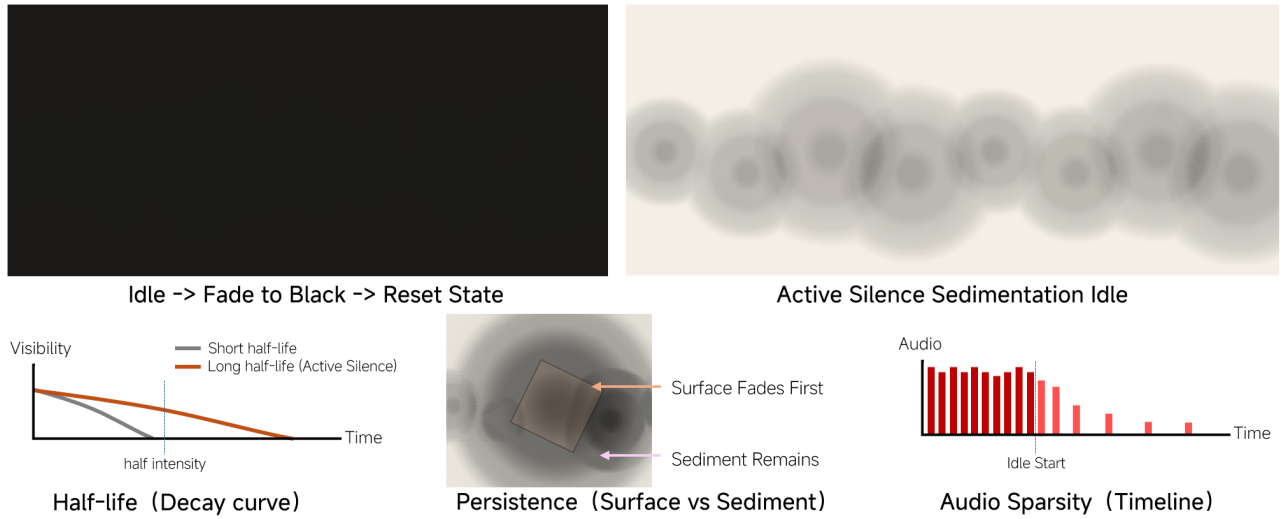


Figure 1: Active Silence replaces reset logic with sedimentation logic and exposes temporal texture as design parameters.

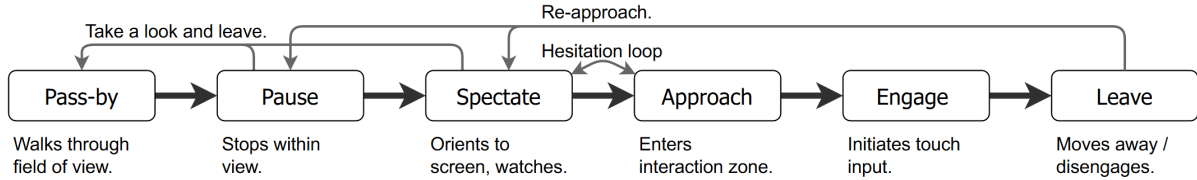


Figure 2: Behavior flow model used for structured observation.

interpreted the quiet/slow idle behavior. We took anonymized notes without recording audio.

4.4 Analysis and Ethics

Observation logs were summarized per condition using descriptive counts and medians. Interview notes were coded thematically, focusing on invitation, pressure, interpretation of traces/silence, and comfort in public. Ethically, the presence proxy was binary and local; no images were stored; signage disclosed sensing. This aligns with known concerns that subtle systems can shape behavior even without users' awareness [2].

In summary, the method is intentionally lightweight but triangulated: observation captures how people move and linger, while interviews capture how they interpret invitation and pressure.

5 Findings

Our findings emphasize qualitative shifts in entry and interpretation. The aim is to characterize how idle policy changes the conditions of engagement, not to claim universal effect sizes.

First, Active Silence increased watchability. In observation, more passersby (21 vs. 7) paused and spectated when the display remained faintly active with lingering traces. The black-screen idle more often read as inactive, reducing the chance that people would discover it as interactive at all.

Second, Active Silence supported gentler transitions from watching to touching. We observed more gradual entries in which people

watched briefly, stepped closer, and then made small exploratory touches. In interviews, participants described the sedimentation state as calm and patient.

Third, our preliminary interviews suggested themes of performance pressure. Under black idle, most visitors (4 out of 5) reported uncertainty about what was expected and whether they needed to do it correctly in public. Under Active Silence, they more often framed interaction as optional: the display was already doing something, and touching merely joined an ongoing process. This aligns with slow-technology accounts in which continuity supports revisitation and low-stakes engagement [12, 13].

Finally, persistence enabled re-approach. When traces remained, some visitors returned to check whether the marks changed or to continue an earlier interaction thread. This suggests that idle continuity can create a sense of temporal ownership without personalization.

To close, the findings point to a coherent mechanism: temporal texture legitimizes spectating and reduces perceived demand, which in turn makes engagement more likely to occur as a gradual social transition.

6 Discussion and Implications

This section deepens the implications by focusing on idle as a social contract, the ethics of calmness, and the practicalities of deploying temporal textures in shared spaces.

6.1 Idle as a Social Contract, Not a Timeout

An idle policy is a public-facing statement about what kind of participation is expected. A black screen can communicate that nothing is happening unless a person initiates, effectively placing the burden of beginning on the visitor. An attract loop can do the opposite, communicating that the display is competing for attention and that passersby are targets. Active Silence proposes a third stance: the display continues at low intensity, making it possible to watch without committing and to join without feeling that one is starting a performance.

This matters because in public space the first barrier is often social rather than technical. People decide whether they are allowed to stop, where they can stand, and whether touching will look awkward. Prior work argues that engagement often progresses from implicit to explicit interaction [14] and that spectatorship supports discovery [10]. Our findings add that the unattended state can either strengthen or weaken that pathway. In practice, the idle state becomes the first tutorial: not by explaining functions, but by expressing pacing and permission.

6.2 Temporal Texture as a Transferable Design Material

Treating idle as temporal texture is useful because it generalizes beyond one aesthetic style. Decay, persistence, and sparsity can be applied to text, images, data visualizations, or playful installations. For example, a community noticeboard might allow posts to fade gradually during quiet periods rather than clearing to a blank template; a museum label might retain subtle traces of prior interactions to suggest that engagement is common and acceptable.

This parameter vocabulary also creates a practical way to iterate with stakeholders. Curators and facilities staff can often discuss whether a space should feel lively or quiet, but they may struggle to specify interaction mechanics. Temporal texture offers intermediate terms: how long should something remain visible, what should remain, and how often should sound occur. This complements evaluation concerns raised for ambient systems by making the design surface explicit and tunable [9, 11].

Temporal textures also introduce failure modes. If visual half-life is too long, sediment can accumulate into visual noise, reducing legibility and making the display appear dirty rather than calm. If it is too short, the system collapses back into a de facto reset and loses continuity. Presence proxies can also misfire (false absence when someone stands still, or false presence due to passersby), producing jittery transitions; in practice, hysteresis and minimum-duration timers can stabilize state changes without increasing sensing granularity.

6.3 Calmness Has Ethics: Quiet Can Still Influence

A calm display is not automatically benign. Subtle systems can shape behavior precisely because they do not announce themselves [2]. Active Silence therefore frames calmness as a responsibility: if a system influences pacing and comfort, it should be legible enough that people can interpret it without feeling manipulated. In our deployment we used binary presence sensing and disclosure signage, and we avoided any inference about identity or emotion.

The broader implication is that designing for calm in public should include an explicit threat model: what is sensed, what is stored, who can access it, and what the system is optimizing for.

There is also a fairness dimension. Displays that require confident, immediate interaction can exclude people who are cautious, unfamiliar with interactive exhibits, or navigating social anxiety. Temporal textures that legitimize spectating may broaden participation by lowering the entry cost, even if they do not maximize throughput. In that sense, Active Silence shifts success criteria from number of touches to quality of approach.

6.4 Energy and Sustainability: Beyond a Binary Choice

Black screens are often justified as energy-saving and less distracting. Our argument is not that displays should remain on, but that energy-aware design can coexist with a calm idle presence. Active Silence can reduce brightness, lower frame rates, and decrease computation during sedimentation, potentially approaching low-power operation while still providing continuity. Psychological amenity is itself a resource in shared spaces: reducing pressure and ambient aggression can improve the experience of a gallery or lobby in ways that are not captured by click-through metrics.

This perspective aligns with work that treats energy awareness as a design space rather than a hidden constraint [7]. For deployments, the practical implication is to negotiate a joint budget: a power budget and an attentional budget. Active Silence is one technique for spending less of both.

To summarize, the discussion positions idle as a deliberate public interaction layer. Temporal texture offers a transferable vocabulary for designing that layer, but it must be deployed with ethical clarity and site-specific tuning.

7 Limitations

This study is limited by the realities of in-the-wild exhibition research. Notably, there is a potential confound in our comparative design: the black screen baseline provides no system feedback, which may lead passersby to assume the display is broken rather than intentionally idle, whereas Active Silence maintains interactive visual cues. Crowd levels and local events can affect engagement independent of idle policy, and our observation is descriptive rather than statistically conclusive [9]. Intercept interviews were brief and few (N=10). Finally, we did not measure well-being directly; claims about reduced pressure reflect self-report and observed approach patterns rather than validated psychological instruments.

These limitations are also pointers for future work: longer deployments across sites, instrumented power measurement, and stronger outcome measures for comfort and attention.

8 Conclusion

Public interactive displays communicate most of the time when nobody is using them. As a preliminary design exploration, Active Silence suggests how idle time might be designed as temporal texture through persistence, decay, and sparsity to support spectatorship, potentially reduce perceived performance pressure, and encourage calm, gradual entry in shared spaces.

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